Parametrically Deformable Contour Models

Lawrence H. Staib* and James S. Duncan†,*

Departments of *Electrical Engineering and †Diagnostic Radiology
Yale University, New Haven, CT 06510

1 Abstract

Segmentation using boundary finding is enhanced both by considering the boundary as a whole and by using model-based shape information. We apply flexible constraints in the form of a probabilistic deformable model to the problem of segmenting natural objects whose diversity and irregularity of shape makes them poorly represented in terms of fixed features or form. The parametric model is based on the elliptic Fourier decomposition of the boundary. The segmentation problem is solved as an optimization problem where the best match between the boundary, as defined by the parameter vector, and the image data is found. Initial experimentation shows good results on a variety of images.

2 Introduction

Segmentation by boundary finding has often been frustrated by poor-contrast boundary regions due to occluding and occluded objects, adverse viewing conditions, and noise. A model-free interpretation is doomed by the underconstrained nature of the problem. Imperfect image data can be augmented with the extrinsic information that a model provides. In order to exploit model-based information to the fullest extent, it should be incorporated explicitly, specifically and early in the analysis. When the model is hidden in algorithm heuristics, it may perform well, but the generality is lost. Some models incorporate generic information such as smoothness or low overall curvature. While this is completely appropriate when no better information is available, the more specific the shape information used, the better. Applying the model too late allows inconsistencies to be committed to by the low-level processing. In addition, the boundary can be profitably considered as a whole because it tends to result in a more overall consistent solution.

This work is aimed at segmenting natural objects whose diversity and irregularity of shape makes them poorly represented in terms of fixed features or form. The objects are expected, however, to have a tendency toward some average shape but there is a continuum of possibilities near that shape. This model domain is somewhere between the extremes of a completely arbitrary object and a fixed template. This tendency can be taken advantage of by its expression in a suitable model.

Other investigators have considered whole-boundary methods that adjust a tentative boundary in order to match to the image. Widrow[1] uses parametrized templates called rubber masks to model objects. The parameters are sizes and relationships between subparts. This approach requires the object to have sufficient structure to be represented in terms of parts. In [2], Schudy uses a spherical harmonic parametrization for boundary finding (in this case, to moving, three-dimensional data). This model provides smoothness, but no explicit shape information. Both of these methods match the model to the image data by searching the parameter space for the best fit.

The boundary finding method of Gritton and Parrish[3] uses a flexible bead chain, where the beads are putative boundary points. The beads are adjusted to neighboring locations to improve the match to the image, in this case, image gradient strength. Prior information is incorporated by the use of an initial boundary and adjusting algorithmic parameters dictating the influence of neighboring beads. In [4], the match metric used is statistical likelihood, resulting in a correlation measure. A similar boundary adjustment scheme is presented to perform the optimization. Kass et al.[5] use energy minimizing snakes that are attracted to image features such as lines and edges while internal spline forces impose a smoothness constraint. The weights of the smoothness and image force terms in the energy functional can be adjusted for different behavior. In addition, they provide a more general and unified treatment of the optimization process by being able to take advantage of standard numerical techniques for partial differential equations.

All of these approaches take an initial estimate of the contour and adjust it to optimize some measure of fit. None of them, except Widrow's, uses information about the shape of the object except implicitly in the initial placement of the contour and generic smoothness properties. In order to use shape information without relying on a decomposition into parts, we can use a generic parametrization with probability distributions defined on the parameters. That is, the parametrization itself will be expressive enough to represent any potential shape, but the associated probability distributions will introduce a bias towards expected or more likely shapes. The spread in the distributions would be due to variability among instances of the object. The distributions for each of the parameters of the model could be estimated from a sample of manually segmented images by decomposing the boundaries into their model parameters and collecting statistics. Alternatively, this method could

CH2752-4/89/0000/0098$01.00 © 1989 IEEE

be run (with manual initialization) on a set of exemplar images.

3 Parametrization

A generic parametrization for closed contours can be defined based on the elliptic Fourier decomposition of the boundary[6,7]. Any closed contour may be represented by two periodic functions of t, where t varies from 0 to 2π, $x(t)$ and $y(t)$, which can in turn be expressed by their Fourier expansions in matrix form as:

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} a_0 \\ c_0 \end{bmatrix} + \sum_{k=1}^{\infty} \begin{bmatrix} a_k & b_k \\ c_k & d_k \end{bmatrix} \begin{bmatrix} \cos kt \\ \sin kt \end{bmatrix} \quad (1)$$

where:

$$\begin{aligned} a_0 &= \frac{1}{2\pi}\int_0^{2\pi} x(t)dt \\ c_0 &= \frac{1}{2\pi}\int_0^{2\pi} y(t)dt \\ a_k &= \frac{1}{\pi}\int_0^{2\pi} x(t)\cos kt\, dt \\ b_k &= \frac{1}{\pi}\int_0^{2\pi} x(t)\sin kt\, dt \\ c_k &= \frac{1}{\pi}\int_0^{2\pi} y(t)\cos kt\, dt \\ d_k &= \frac{1}{\pi}\int_0^{2\pi} y(t)\sin kt\, dt \end{aligned}$$

This particular version of Fourier boundary representation has a number of advantages. Both the decomposition and the reconstruction can be calculated explicitly and efficiently due to the fast Fourier transform. Since the x and y functions are periodic and continuous, any truncation of the expansion will reconstruct to a closed boundary (unlike some other Fourier representations[8]). The parameters of the decomposition are also more understandable in terms of a geometric interpretation. Invariance to rotation, scale, translation and starting point can also be easily achieved.

In equation 1, the first two coefficients, a_0 and c_0, determine the overall translation of the shape. Each term in the summation can be shown to define an ellipse[6]. The matrix therefore determines the characteristics of the ellipse. The contour can be viewed as being decomposed into a sum of rotating phasors, each individually defining an ellipse, and rotating with a speed proportional to their harmonic number, k. This can be seen in figure 1 where the straight lines represent these phasors for a contour constructed from three ellipses (harmonics). The ellipses, denoted by their center points, c_0, c_1 and c_2 are shown at three different times. The last point, c_3, traces out the curve.

The properties of these ellipses can be derived from the raw matrix elements. Each ellipse can be described by the lengths of its major and minor axes, its rotation (the angle from the x-axis of the major axis, defined from $-\pi/2$ to $\pi/2$) and its phase shift (the phase from the major axis to the position of $t = 0$, defined from $-\pi$ to π).

In order to determine the ellipse parameters, first consider an ellipse with its major axis aligned with the x-axis with no

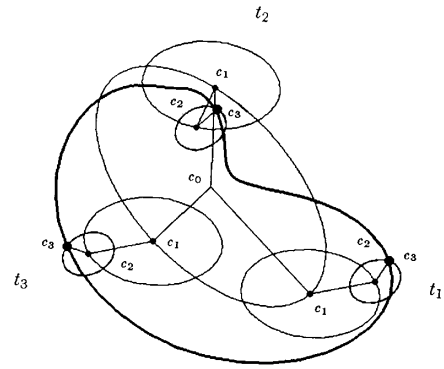

Figure 1: Contour constructed from three ellipses with components shown at three different times.

phase shift. This ellipse would be represented as:

$$\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \quad (2)$$

where A and B are the major and minor semi-axis lengths, respectively. The phasor moves counterclockwise for B positive, clockwise for B negative. A rotation of this ellipse by θ and shift by ϕ can be written as:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix} \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix} \quad (3)$$

This is equal to:

$$\begin{bmatrix} A\cos\theta\cos\phi - B\sin\theta\sin\phi & -A\cos\theta\sin\phi - B\sin\theta\cos\phi \\ A\sin\theta\cos\phi + B\cos\theta\sin\phi & -A\sin\theta\sin\phi + B\cos\theta\cos\phi \end{bmatrix}$$

This represents a general ellipse. To find the ellipse parameters given the values of these matrix elements, solve the following four equations for A, B, θ and ϕ.

$$\begin{aligned} a &= +A\cos\theta\cos\phi - B\sin\theta\sin\phi \\ b &= -A\cos\theta\sin\phi - B\sin\theta\cos\phi \\ c &= +A\sin\theta\cos\phi + B\cos\theta\sin\phi \\ d &= -A\sin\theta\sin\phi + B\cos\theta\cos\phi \end{aligned} \quad (4)$$

This results in:

$$\begin{aligned} A^2 &= \frac{i + \sqrt{\alpha^2 - 4\beta^2}}{2} \\ B^2 &= \frac{2\beta^2}{i + \sqrt{\alpha^2 - 4\beta^2}} \end{aligned} \quad (5)$$

where

$$\begin{aligned} \alpha &= a^2 + b^2 + c^2 + d^2 \\ \beta &= ad - bc \end{aligned} \quad (6)$$

and

$$\begin{aligned}\theta &= \tan^{-1}\frac{Ac+Bb}{Aa-Bd}\\ \phi &= \tan^{-1}\frac{Ba-Ad}{Ac+Bb}\end{aligned} \tag{7}$$

By taking A to be positive and B to agree in sign with j, we get a consistent sign convention.

The phase shift and rotation are then best converted to relative values. A rotation of θ on the first ellipse results in a rotation of θ on the kth ellipse and a phase shift of ϕ on the first ellipse results in a phase shift of $k\phi$ on the kth ellipse. In order to remove the cumulative effects of the rotation and shift, we get the following formulas for converting between absolute rotation and shift (θ and ϕ) and rotation and shift relative to the preceding ellipse (θ' and ϕ').

$$\begin{aligned}\theta'_k &= \theta_k - \theta_{k-1}\\ \theta_k &= \sum_{l=1}^{k}\theta'_l\\ \phi'_k &= \phi_k - \sum_{l=1}^{k-1}(k-l+1)\phi_l\\ \phi_k &= \sum_{l=1}^{k}(k-l+1)\phi'_l\end{aligned} \tag{8}$$

In this way, one parameter, θ'_1, determines the overall rotation and allows the phase shift of the first ellipse to be set to zero, since it is arbitrary. The axes lengths can be normalized to the first major axis in order to isolate a single parameter that determines the overall scale:

$$\begin{aligned}A'_k &= \frac{A_k}{A_1} \quad \text{for} \quad k \neq 1\\ B'_k &= \frac{B_k}{A_1}\end{aligned} \tag{9}$$

Thus, we have explicit equations for the conversion between the raw coefficients of the Fourier expansion and the more readily understandable ellipse properties, A', B', ϕ' and θ'.

The number of harmonics necessary for an accurate representation increases with the high frequency content of the contour. Many biological forms are relatively smooth and unconvoluted and thus can be well represented by a small number of components. Limiting the number of harmonics has the advantage of inducing a smoothing effect on the segmentation. Too many harmonics will unnecessarily increase the computational complexity. If high order harmonics were included, the probability distributions associated with their axis lengths would be clustered about zero, thus minimizing their contribution. For the examples we have been considering, between four and six terms of the expansion have been sufficient.

4 Optimization

To match the model to the image, we use correlation as the match metric. We could correlate the image data, I, directly with a template, formed from the parameter vector, $\mathbf{p}$, consisting of the ellipse property coefficients, A'_k, B'_k, ϕ'_k, θ'_k derived from the Fourier decomposition as shown in equations 8 and 9. However, since the boundary of the object is the goal of this analysis, we are better off computing the correlation between the gradient of the image, as a measure of edge strength, and an edge template, $t_{\mathbf{p}}(x,y)$, which has support only along the contour defined by $\mathbf{p}$:

$$C(\mathbf{p}) = \nabla I \star\star t_{\mathbf{p}} = \iint \nabla I(x,y)\, t_{\mathbf{p}}(x,y)\, dx\, dy \tag{10}$$

This will both reduce the computation not only of the correlation itself but of the template generation, and improve the signal-to-noise ratio. Since the template represents the boundary, this can be written as a line integral:

$$C(\mathbf{p}) = \int_{v_{\mathbf{p}}} \nabla I(x,y)\, ds \tag{11}$$

where the path of integration is represented parametrically by the template boundary position $v_{\mathbf{p}} = (x_{\mathbf{p}}(s), y_{\mathbf{p}}(s))$

The gray level gradient can be calculated by first smoothing with a Gaussian to reduce the effect of noise. Then calculate a finite difference approximation to the x and y components of the gradient. This will result in a smoothly varying gradient, relatively undegraded by noise that will guide the optimization.

Under the assumption of zero-mean, independent Gaussian noise corrupting an image, the correlation of an image, I, with a template t is proportional to $\log P(I|t)$, that is, the logarithm of the probability of the observed image, given that the object is present[9]. For our purposes, we have the gradient of the image correlated with a template defined by a parameter vector $\mathbf{p}$, giving $\log P(\nabla I|\mathbf{p})$. The gradient is calculated from a neighborhood of pixels and therefore the values are correlated. However, as a useful approximation, we can consider the assumption valid. To find the maximum a posteriori estimate of the parameter vector $\mathbf{p}$, we take

$$\max_{\mathbf{p}} P(\nabla I|\mathbf{p})P(\mathbf{p}) \tag{12}$$

where $P(\mathbf{p})$ is the prior probability for the parameter vector. This is equivalent, due to the monotonicity of the logarithm, to

$$\begin{aligned}\max_{\mathbf{p}} \log\left(P(\nabla I|\mathbf{p})P(\mathbf{p})\right) &= \max_{\mathbf{p}} \log P(\nabla I|\mathbf{p}) + \log P(\mathbf{p})\\ &= \max_{\mathbf{p}} C(\mathbf{p}) + \log P(\mathbf{p})\end{aligned} \tag{13}$$

An exhaustive search for this optimum is impractical because of the high dimensionality of the parameter space. Instead, we use a local multi-dimensional optimization starting at the maximum of the a priori distributions, with $C(\mathbf{p})+\log P(\mathbf{p})$ as the objective function. The parameter distributions generally provide enough of a guide to make a local optimization sufficient. This correlation gives an overall measure of confidence in the final boundary and can also be calculated locally along the boundary to provide a regional estimate. This procedure, in effect, finds the local maximum of edge strength conforming to the parametrization.

We have numerically optimized this measure using Powell's direction set method[10]. This method has the advantage

of not requiring derivative information. The gradient of our objective is expensive to compute since it can not be analytically evaluated but only numerically approximated. Powell's method starts by minimizing along individual parameter directions, then looks for promising combinations of parameters to optimize along based on its experience. It may be possible, however, to obtain faster convergence using conjugate gradient or quasi-Newton methods despite the gradient computation.

Gradient strength is just one feature that could be used for the correlation. In principle, any measure that is indicative of boundary strength could be used. In particular, figure 4 shows the use of a measure of line strength as an indication of a boundary.

As an extension of the basic formulation, a multi-resolution strategy could be used to improve the convergence properties. Different amounts of spatial smoothing could be applied to the image. Then, the optimization could be run at increasing levels of resolution while also increasing the number of harmonics. Also, additional information from adjacent (temporal or spatial) frames and comparable views in other modalities could add reliability. This would involve constraining the the parameter vector to smoothly vary by frame.

5 Results

The algorithm has been tried on a variety of deformable object boundary finding problems. The examples shown here include a synthetic image and two medical images.

The image shown in figure 2 is a simple synthetic Mondrian image where the target object (the darkest) is partially occluded by one object on the right and overlaps another. The extraneous objects are potential sources of choice or confusion because alternate boundaries are plausible especially at the right side of the target and at the center of the top boundary. The initial curve superimposed is an artificially generated starting boundary defined by a parameter vector $\mathbf{p}$ derived from the mean curve of a set of manually traced contours. The starting boundary only roughly agrees with the target in terms of shape and location. The final curve accurately delineates the target. The curve correctly avoids both the overlapped and the occluding object both because of the limit on the number of harmonics (four) and the bias due to the probability distributions on the parameters.

In cardiac imaging, it is desired to isolate the boundary of the left ventricle of the heart (LV) in order to quantify cardiac wall motion and shape. In two-dimensional Echocardiography (2DE) the boundary to be identified is suggested in the image more by a subjective grouping of intensity information such as discussed by [11], rather than by complete, closed groups of edge gradients. As seen in the 2DE apical long axis view of the LV at end-diastole shown in figure 3, the image is largely made up of organized blobs of signal intensity. The original gray level of the image and the magnitude of the intensity gradient are shown. The initial contour is generated from a sample of manually traced contours. Note that although this is almost a plausible shape for this boundary, many of the strongest gradients such as those near the lower left part of the contour (the posterobasal wall) are completely missed. The final converged solution of the algorithm attains a more acceptable shape in this region, while the remainder of the contour still adheres to the most plausible portions in the other regions. Note the extension of the apex of the perceived contour off the end of the upper leftmost portion of the image. Apical dropout of this sort often occurs in 2DE studies taken from this view. Thus, this complete boundary model approach may have further merit in estimating the shape of the complete LV even in regions where it is entirely missing.

The algorithm has also been tried on magnetic resonance images of the corpus callosum in the human brain. The results are shown in figure 4. The size and shape of the corpus callosum are important in the study of its relationship to cognitive function[12]. In these images, the corpus callosum is separated from the rest of the brain by a dark line. Thus, a measure that responded to lines was used instead of the gray-level gradient. In this case, we used the positive magnitude of the Laplacian of the Gaussian, which responds to both lines and edges. The final contour succeeds in delineating the structure properly. Because of the narrow, curved shape of the corpus callosum, six harmonics are needed to accurately represent it.

This method appears to work well delineating structures, and to be relatively insensitive to the problems of broken boundaries and spurious edges from nearby objects. Initial experimentation shows good results on a variety of images. The flexibility of the model both in terms of its probabilistic nature and the parametric representation make this an attractive method for boundary finding.

6 Acknowledgements

This work is supported in part by NIH Grant T15LM07056 from the National Library of Medicine.

References

[1] Bernard Widrow. The "rubber mask" technique – I. and II. *Pattern Recognition*, 5:175–211, 1973.

[2] Robert B. Schudy. *Harmonic surfaces and parametric image operators: Their use in locating the moving endocardial surface from three-dimensional cardiac ultrasound data.* Computer Science Technical Report 112, University of Rochester, Rochester, New York, March 1981.

[3] C. W. K. Gritton and Jr. Edward A. Parrish. Boundary location from an initial plan: The bead chain algorithm. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 5(1):8–13, January 1983.

[4] David B. Cooper. Maximum likelihood estimation of markov-process blob boundaries in noisy images. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 1(4):372–384, October 1979.

[5] Michael Kass, Andrew Witkin, and Demetri Terzopoulos. Snakes: Active contour models. *International Journal of Computer Vision*, 1(4):321–331, 1988.

[6] Frank P. Kuhl and Charles R. Giardina. Elliptic fourier features of a closed contour. *Computer Graphics and Image Processing*, 18:236–258, 1982.

[7] Chun-Shin Lin and Chia-Lin Hwang. New forms of shape invariants from elliptic fourier descriptors. *Pattern Recognition*, 20(5):535–545, 1987.

[8] E. Persoon and K. S. Fu. Shape discrimination using fourier descriptors. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(3):388–397, May 1986.

[9] Azriel Rosenfeld and Avinash C. Kak. *Digital Picture Processing*. Volume 2, Academic Press, Orlando, Florida, second edition, 1982.

[10] William H. Press, Brian P. Flannery, Saul A. Teukolsky, and William T. Vetterling. *Numerical Recipes*. Cambridge University Press, Cambridge, 1986.

[11] David Marr. *Vision*. W. H. Freeman and Company, New York, 1982.

[12] Sandra F. Witelson and Debra L. Kigar. Anatomical development of the corpus callosum in humans: A review with reference to sex and cognition. In D. L. Molfese and S. J. Segalowitz, editors, *Developmental implications of brain lateralization*, Guilford Press, New York, 1988.

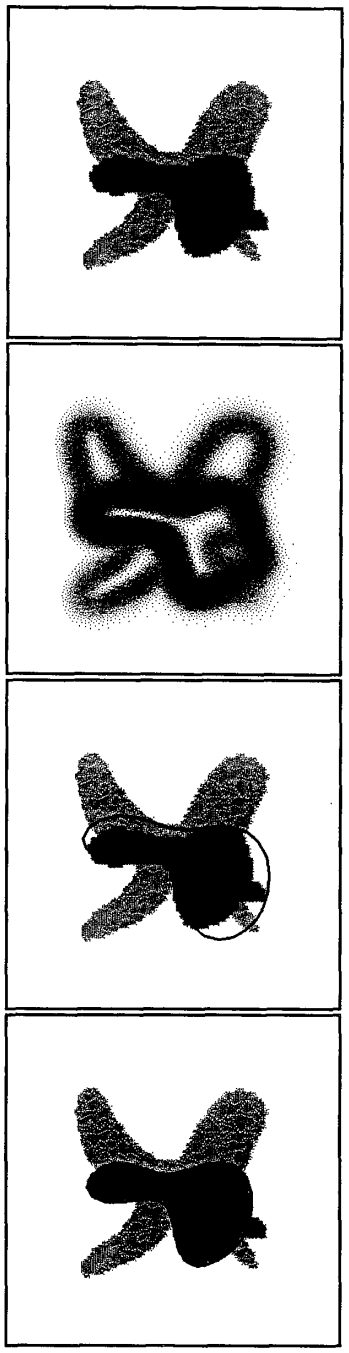

Figure 2: From top to bottom: Synthetic image; Gray level gradient magnitude; Initial contour; Final contour.

Figure 3: From top to bottom: Echocardiogram, long axis view; Gray level gradient magnitude; Initial contour; Final contour.

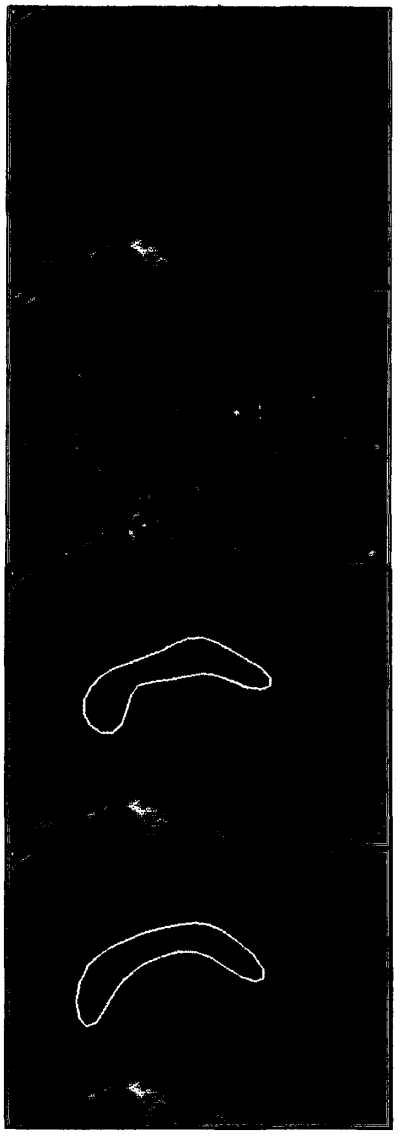

Figure 4: From top to bottom: NMR mid-brain sagittal image; Positive magnitude of the Laplacian of the Gaussian; Initial contour; Final contour.